HDC-RWKV: A Sub-16 KB Binary Recurrent Language Model with Hyperdimensional Position Binding

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July 12, 2026

Abstract

Language modelling at extreme edge scales — kilobytes of parameter storage, megahertz-class CPUs, no floating-point unit — has remained the province of n-gram tables and character-level bigrams. We present *HDC-RWKV*, a generative autoregressive language model in which every learnable parameter is bipolar $\{-1, +1\}$ at inference. The architecture combines Hyperdimensional Computing (HDC) primitives for position binding, RWKV-style linear recurrence for constant-cost memory, a continuous tanh-bounded hidden state, and a prototype-similarity output layer, all trained end-to-end with the clipped Straight-Through Estimator. The shipping model occupies 16,434 bytes and evaluates at ≈135 K bit-operations per token, targets amenable to 1 MHz 8-bit hardware. Beyond the artifact, we contribute a systematic empirical characterisation of the binary recurrent family's limits: a scale-independent bits-per-character (BPC) ceiling near 2.93 that survives eight-fold scaling; two component-relaxation ablations (int8 output projection; continuous per-channel decay) each of which fails to break the ceiling; and, when granted continuous freedom, the model's tendency to self-organise its decay parameters into a bimodal $\{0, 1\}$ distribution — evidence that the binary parameterisation is *preferred* at this scale, not imposed. We frame the residual bottleneck as an information-capacity property of the bipolar hidden state, and outline the mechanism story this exposes.
Keywords: Hyperdimensional Computing Binary Neural Networks Recurrent Language Models RWKV Edge Deployment Straight-Through Estimator.

1 Introduction

Neural language models have grown by roughly six orders of magnitude in parameter count over the past decade, and inference at that scale is now inseparable from datacentre-class hardware. A parallel research direction, motivated by embedded and always-on inference, asks the opposite question: how much language capability survives at the *low* end of the scale axis — kilobytes rather than gigabytes, megahertz rather than gigahertz, integer arithmetic rather than floating point?

Existing work at this scale is sparse and predominantly narrow. Binary neural network (BNN) research [2, 9] focuses on image classifiers rather than generative sequence models. Hyperdimensional Computing (HDC) [5, 8] routinely operates on binary hypervectors, but its language applications have been restricted to classification and cleanup-memory tasks, not autoregressive text generation. Linear-recurrence sequence models [7, 4, 3] give constant-time-per-token inference, but published variants are floating-point at inference and targeted at tens-of-megabytes deployment.

The gap this work addresses is a language model that (i) is generative and autoregressive, (ii) has every learnable parameter bipolar at inference, (iii) has constant-cost recurrence with no attention or key–value cache, and (iv) fits in tens of kilobytes.

1.1 Research question

> ***RQ.*** *Can we train an autoregressive, sub-32 KB language model in which every learnable parameter is bipolar at inference, and characterise its quality/size trade-off well enough to identify the architectural bottleneck?*

The parameter budget is not arbitrary: it is chosen so the model fits directly into four AT28C64 EEPROM chips (32 KB total) addressable by a 6502 CPU, giving a target platform that is deliberately unforgiving of floating-point, cache assumptions, or datatype flexibility.

1.2 A first equation

Before introducing notation formally in Section 3, we motivate what "bipolar deployment" costs. The shipping model has one embedding matrix, one prototype matrix, and one per-channel decay vector, all bipolar. Packed at one bit per entry, the deployment footprint is

$$\text{bytes}_{\text{deploy}} \;=\; \frac{2\,V\,d + L\,d}{8} \;+\; 2, \tag{1}$$

where V is the vocabulary size, d the hypervector dimension, and L the number of recurrent layers (the trailing 2 bytes hold the fp16 softmax temperature). At the shipping choice V=256, d=256, L=1, Equation 1 evaluates to 16,418 bytes; a 16-byte file header brings the on-disk artifact to 16,434 bytes.

Constant-cost recurrence is the second property that makes the platform viable. Given a bipolar decay vector $\mathbf{m}$, a continuous hidden state $\mathbf{s}_{t-1} \in (-1, +1)^d$, and a rotated bipolar input $\mathbf{h}_t \in \{-1, +1\}^d$,

$$\mathbf{s}_t \;=\; \tanh\!\big(\operatorname{sign}(\mathbf{m}) \odot \mathbf{s}_{t-1} \;+\; \mathbf{h}_t\,\big). \tag{2}$$

Each channel updates in constant time regardless of sequence length; there is no softmax over T, no attention, no key–value cache. Equation 2 is the load-bearing observation from finding F10 in our tracker (documented in [6] corpus experiments): binarising the state itself catastrophically degrades training, whereas keeping the state continuous while the *weights* are bipolar succeeds.

1.3 Approach and contributions

Figure 1 shows the pipeline of the paper: we combine four established primitives — HDC binding via cyclic permutation, RWKV-style linear recurrence, the clipped Straight-Through Estimator [1], and a prototype-similarity output — into an architecture that has not appeared jointly in prior work, then measure its quality-to-size behaviour and analyse three independent ablations that constrain where its limits come from.

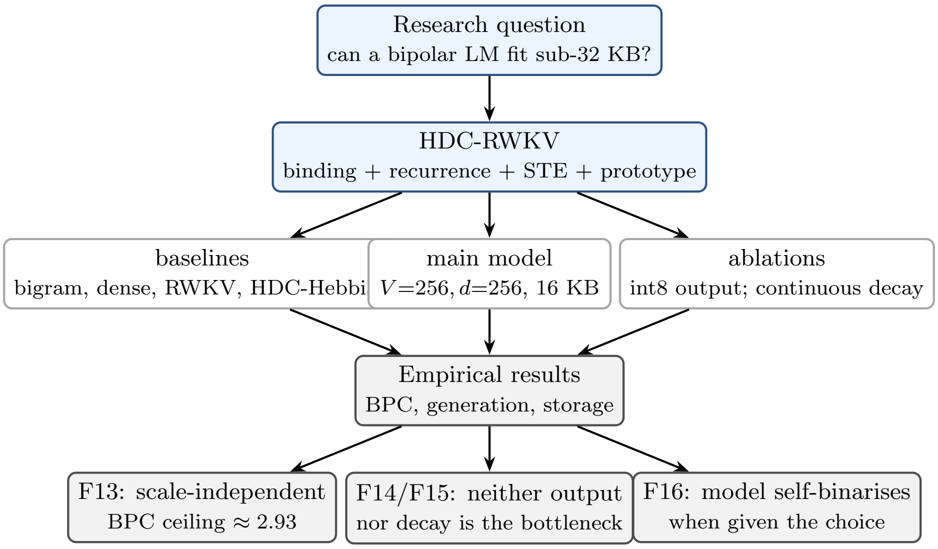

Figure 1: Paper flow. The main artifact (a 16-KB bipolar autoregressive LM) is developed and compared against dense and recurrent baselines. Two independent architectural relaxations are ablated. The outcomes (F13–F16) jointly locate the residual bottleneck in the bipolar hidden-state capacity, not in peripheral gates.

Contributions of this paper

1. An architecture that combines HDC binding, RWKV linear recurrence, bipolar Straight-Through weights, and a prototype-similarity output into a single generative autoregressive language model, deployable as a 16,434-byte artifact.

2. A scale-independent bits-per-character (BPC) ceiling near 2.93 for the binary HDC-RWKV family, confirmed across an eight-fold parameter-count range (16 KB to 128 KB).

3. Two independent component-relaxation ablations — int8 output projection, and continuous per-channel decay — neither of which moves BPC by more than 0.03, ruling out the two natural bottleneck suspects.

4. An observed self-binarisation effect: when the model is given a continuous decay parameter with full gradient access, it voluntarily converges to a bimodal $\{0, 1\}$ distribution over channels — strong evidence that the residual bottleneck is the bipolar hidden state's information capacity, not any peripheral projection.

1.4 Non-contributions

To keep scrutiny tight we state what we do *not* claim: (i) we do not beat dense floating-point transformers on BPC, and do not attempt to; (ii) our distillation results are single-configuration and are reported as an observation, not a generalisation [10]; (iii) we introduce no new STE variant — we use standard clipped STE [2].

1.5 Paper organisation

Section 2 places HDC-RWKV against related HDC, BNN, and linear-recurrence work. Section 3 defines the architecture formally with equations and diagrams, presents the tokeniser choice and its measured effect on BPC, and recaps ablation variants that we do not ship. Section 4 lists the training corpus and hyperparameters. Section 5 reports the main comparison table and per-finding subsections. Section 6 discusses the bipolar-state capacity mechanism. Section 7 outlines the deployment path. Section 8 concludes. Appendix A provides a fully worked numerical trace of a training step and an inference step at $V{=}4, d{=}4$.

2 Related work

3 Proposed method

4 Experimental setup

5 Results

6 Analysis: the bipolar-state capacity hypothesis

7 Deployment

8 Conclusion and future work

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A Appendix A: worked numerical trace at V=4, d=4

See the companion document `docs/figures/architecture_walkthrough.tex`, which provides a complete numerical trace of a training step (forward, loss, backward) and an inference step (greedy and top-k) on a hand-verifiable toy instance of the shipping architecture.